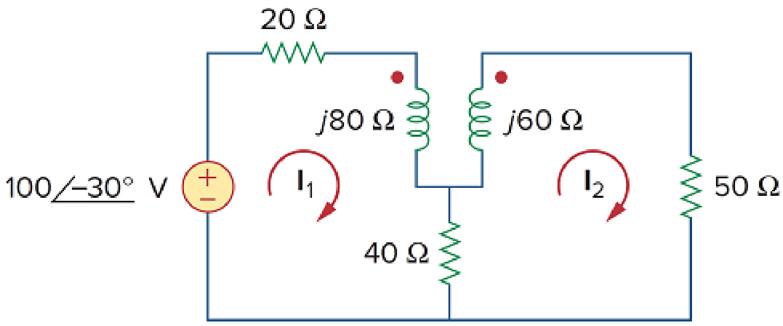
# Chapter 13, Problem 78P

## Problem

Use *PSpice* or *MultiSim* to determine the mesh currents in the circuit of Fig. 13.140. Take $\omega$ = 1 rad/s. Use $k$ = 0.5 when solving this problem.

**Figure 13.140**




## Step-by-step solution

**Step 1** of 7

Refer to Figure 13.140 from the text book.

Coupling coefficient is $k = 0.5$.

Write the formula for coupling coefficient.

$$k = \frac{X_M}{\sqrt{X_{L1}X_{L2}}}$$
$$0.5 = \frac{X_M}{\sqrt{(80)(60)}}$$
$$X_M = 0.5\sqrt{(80)(60)}$$
$$X_M = 34.64\ \Omega$$

**Step 2** of 7

Apply Kirchhoff's voltage law for loop $\mathbf{I}_1$.

$$-100\angle -30^\circ + 20\mathbf{I}_1 + j80\mathbf{I}_1 - j34.64\mathbf{I}_2 + 40(\mathbf{I}_1 - \mathbf{I}_2) = 0$$

$$(60 + j80)\mathbf{I}_1 - (40 + j34.64)\mathbf{I}_2 = 100\angle -30^\circ \quad \text{......} \ (1)$$

Apply Kirchhoff's voltage law for loop $\mathbf{I}_2$.

$$40(\mathbf{I}_2 - \mathbf{I}_1) + j60\mathbf{I}_2 - j34.64\mathbf{I}_1 + 50\mathbf{I}_2 = 0$$
$$-(40 + j34.64)\mathbf{I}_1 + (90 + j60)\mathbf{I}_2 = 0$$
$$(40 + j34.64)\mathbf{I}_1 = (90 + j60)\mathbf{I}_2$$
$$\mathbf{I}_1 = \left(\frac{90 + j60}{40 + j34.64}\right)\mathbf{I}_2$$

$$\mathbf{I}_1 = (2.028 - j0.256)\mathbf{I}_2 \quad \text{......} \ (2)$$

**Step 3** of 7

Substitute $(2.028 - j0.256)\mathbf{I}_2$ for $\mathbf{I}_1$ in equation (1).

$$(60 + j80)(2.028 - j0.256)\mathbf{I}_2 - (40 + j34.64)\mathbf{I}_2 = 100\angle -30^\circ$$
$$(142.16 + j146.88)\mathbf{I}_2 - (40 + j34.64)\mathbf{I}_2 = 100\angle -30^\circ$$
$$(102.16 + j112.24)\mathbf{I}_2 = 100\angle -30^\circ$$
$$(151.77\angle 47.69^\circ)\mathbf{I}_2 = 100\angle -30^\circ$$

$$\begin{aligned}\mathbf{I}_2 &= \frac{100\angle -30^\circ}{151.77\angle 47.69^\circ} \\ &= 0.658\angle -77.69^\circ \text{ A}\end{aligned}$$

Therefore, the mesh current $\mathbf{I}_2$ is $\boxed{0.658\angle -77.69^\circ \text{ A}}$.

Substitute $0.608\angle -71.85^\circ \text{ A}$ for $\mathbf{I}_2$ in equation (2).

$$\begin{aligned}\mathbf{I}_1 &= (2.028 - j0.256)(0.658\angle -77.69^\circ) \\ &= (2.044\angle -7.19)(0.658\angle -77.69^\circ) \\ &= 1.345\angle -84.88^\circ \text{ A}\end{aligned}$$

Therefore, the mesh current $\mathbf{I}_1$ is $\boxed{1.345\angle -84.88^\circ \text{ A}}$.

**Step 4** of 7

Calculate the mesh currents using PSpice.

First calculate the inductors values.

$$\omega L_1 = 80$$

$$\begin{aligned}L_1 &= \frac{80}{\omega} \\ &= \frac{80}{1} \\ &= 80 \text{ H}\end{aligned}$$

And,

$$\omega L_2 = 60$$

$$\begin{aligned}L_2 &= \frac{60}{\omega} \\ &= \frac{60}{1} \\ &= 60 \text{ H}\end{aligned}$$

**Step 5** of 7

Draw the schematic diagram of Figure 13.140.

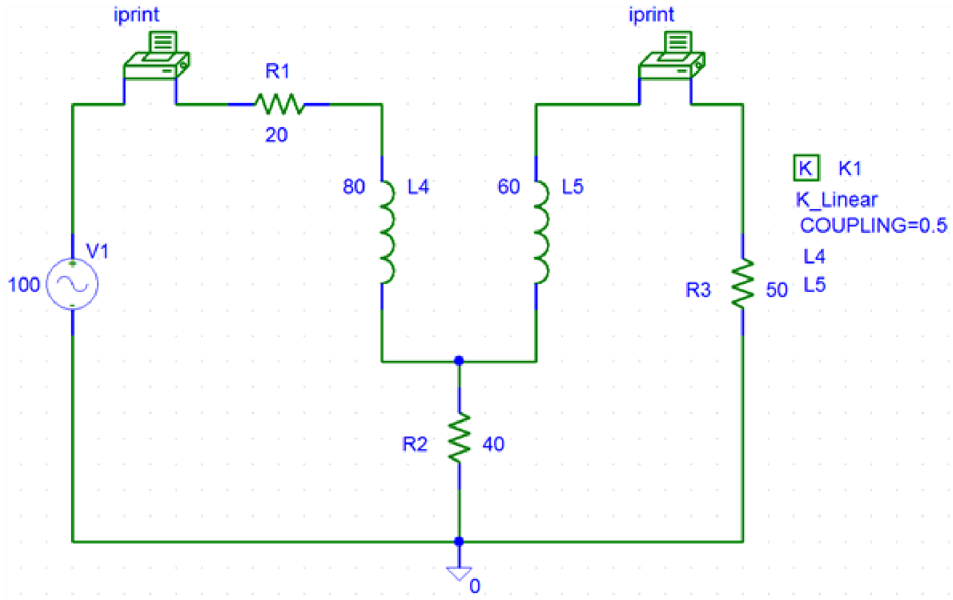



Figure 1

**Step 6** of 7

To handle the coupling inductors, we use K-LINEAR provided in the analog library and set the following attributes(by double clicking on the symbol K in the box):

K K1

K-LINEAR

L1 = L4

L2 = L5

COUPLING = 0.5

**Step 7** of 7

Double click on VAC voltage source and set ACMAG = 100 and ACPHASE = – 30, and double click on IPRINT and set AC = yes, MAG = yes and PHASE = yes.

Next select **analysis /setup/AC sweep** and enter total pts = 1, start frequency = 0.1592 Hz and Final frequency = 0.1592 Hz. After saving the schematic, select analysis/simulate to simulate it. The output file includes:

FREQ IM(V_PRINT1) IP(V_PRINT1)

1.592E-01 1.347E+00 -8.489E+01

FREQ IM(V_PRINT2) IP(V_PRINT2)

1.592E-01 6.588E-01 -7.769E+01

Therefore, the mesh currents $\mathbf{I}_1$ and $\mathbf{I}_2$ are $\boxed{1.345\angle -84.88^\circ \text{ A and } 0.658\angle -77.69^\circ \text{ A}}$.